PHY4605 Computational Methods in Physics
Learning Note | Week 5

Root Finding: Locate a physical solution and verify the residual

Turn a physical target condition into a residual, then find and validate one meaningful root.

Physical question. A projectile is launched from level ground with speed $20\,\mathrm{m\,s^{-1}}$. What low launch angle makes the range equal to a 35 m target distance?

Learning outcomes

1. rewrite a physical target condition as a residual (the signed model-minus-target mismatch) equation $f(x) = 0$;
2. identify a sign-changing bracket and trace starter bisection and Newton updates; and
3. validate a computed root by substituting it back into the residual and interpreting its unit.

How to use this note

Read the physical question and residual definition before the MATLAB code. Pause at each prediction, bracket, and validation checkpoint before running the lecture demonstration. Keep the unknown, its unit, the algorithm, and the physical check connected.

1. A root is a physical condition, not just a number

A root-finding problem begins with an original physical equation and a target condition. We do not start by asking MATLAB to "solve for x" in the abstract. We first ask what physical output should equal what target value.

For Week 5, use the level-ground projectile range model

$$R(\theta) = \frac{v_0^2}{g}\sin(2\theta),$$

where θ is the launch angle, $v_0 = 20\,\mathrm{m\,s^{-1}}$, and $g = 9.81\,\mathrm{m\,s^{-2}}$. The target condition is

$$R(\theta) = 35\,\mathrm{m}.$$

The residual is the signed difference between the model output and its target. It tells us how far, and in which direction, the model is from the target. For this problem, the residual is

$$f(\theta) = R(\theta) - 35.$$

A root is an angle where $f(\theta) = 0$, so the model range equals the target range.

Quantity	MATLAB name	Meaning and unit
v_0	`v0_mps`	launch speed, m/s
g	`g_mps2`	gravitational acceleration, $\mathrm{m/s^2}$
θ	`theta_deg`	launch angle, degree; unknown root
R	`range_m`	model range, m
$f(\theta)$	`residual_m`	range residual, m

A sign-changing bracket is an interval whose endpoint residuals have opposite signs. This shows that the residual crosses zero between them, so the interval contains a candidate root.

2. Residual signs have physical meaning

Residual sign	Physical meaning
$f(\theta) < 0$	model range is shorter than the target: the projectile undershoots
$f(\theta) = 0$	model range equals the target: the angle is a root
$f(\theta) > 0$	model range is longer than the target: the projectile overshoots

In plain language, an undershoot falls short of the target and an overshoot goes beyond it. The sign of the residual records which situation the model predicts.

For the values used in this example, a low-angle bracket from 20° to 40° changes sign:

$$f(20^\circ) \approx -8.79\,\text{m}, \qquad f(40^\circ) \approx +5.16\,\text{m}.$$

This does not prove that there is only one projectile angle. It gives the required evidence for one bracketed low-angle root.

Prediction checkpoint. The low-angle root should lie between 20° and 40°. If the residual is still negative, the model undershoots; if it is positive, the model overshoots.

3. See the root on a residual graph

A residual graph plots the mismatch, not the original range. The root is where the residual curve crosses the zero line.

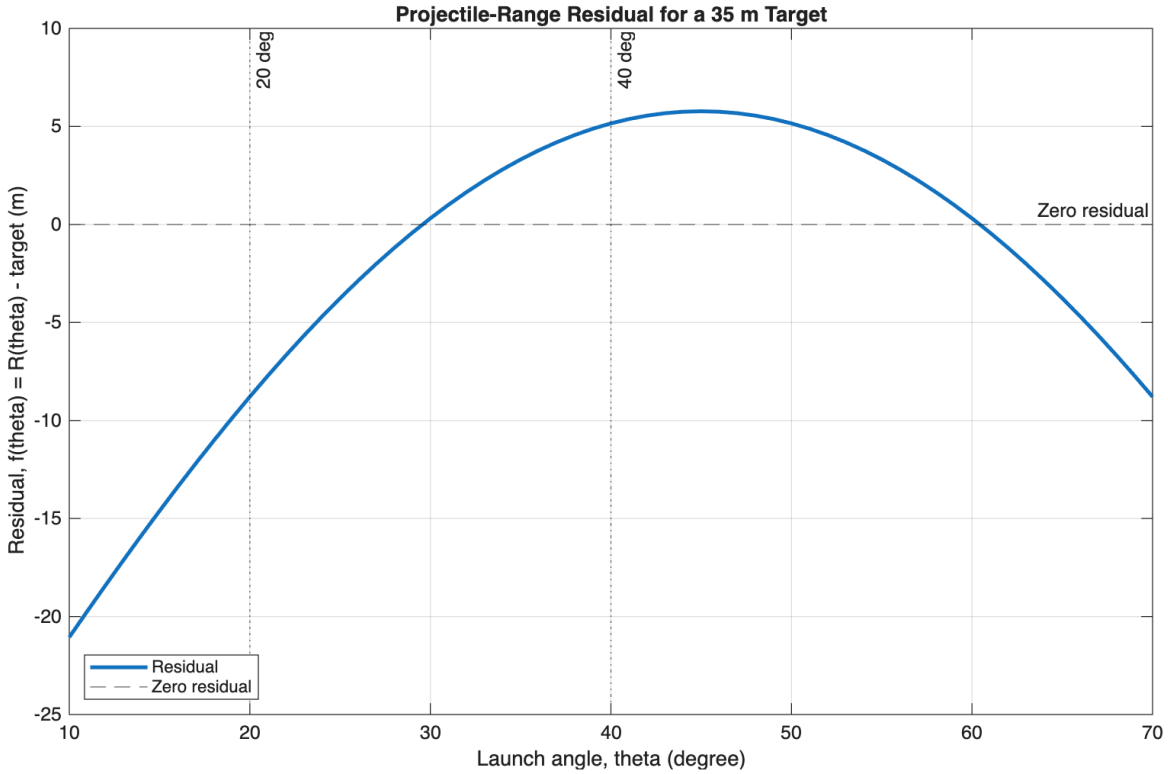


Figure 1: MATLAB-generated residual graph for the projectile model used in this example. The selected low-angle bracket from 20° to 40° contains a sign change.

4. Pseudocode for bisection

Bisection is a safe bracketed method when the residual signs at the interval endpoints are opposite.

1. choose left and right endpoints with opposite residual signs;
2. compute the midpoint;
3. evaluate the residual at the midpoint;
4. if the midpoint residual is small enough, stop;
5. otherwise keep the half interval whose endpoints still have opposite signs; and
6. repeat until the supplied stopping rule is satisfied.

Required stopping rule. In Week 5 required work, stop when $|f(\theta)| < 10^{-3}$ m. Do not compare several stopping rules unless the task is explicitly marked as an optional extension.

5. Starter bisection code in MATLAB

Listing 1: Starter MATLAB implementation of the bracketed bisection update

```
v0_mps = 20;
g_mps2 = 9.81;
target_m = 35;
tolerance_m = 1e-3;
left_deg = 20;
right_deg = 40;
for iteration = 1:40
    mid_deg = (left_deg + right_deg)/2;
    f_mid_m = (v0_mps^2/g_mps2)*sind(2*mid_deg) - target_m;
    if abs(f_mid_m) < tolerance_m
        break
    end
    f_left_m = (v0_mps^2/g_mps2)*sind(2*left_deg) - target_m;
    if f_left_m*f_mid_m < 0
        right_deg = mid_deg;
    else
        left_deg = mid_deg;
    end
end
root_deg = mid_deg;
```

The midpoint is not a magic guess. It is the centre of the current bracket. The update keeps the side of the interval that still contains a sign change.

6. Newton's method as a tangent update

Newton's method uses a local tangent line instead of an interval half. In words:

$$\text{new angle} = \text{current angle} - \frac{\text{current residual}}{\text{current residual slope}}.$$

For the projectile residual with angle in degrees,

$$\frac{df}{d\theta} = \frac{v_0^2}{g}\cos(2\theta)\frac{2\pi}{180}.$$

The factor $2\pi/180$ appears because the update uses degrees while calculus derivatives normally use radians. The slope unit is metres per degree.

Listing 2: Starter MATLAB implementation of Newton's angle update

```
theta_deg = 25;
for iteration = 1:40
    f_current_m = (v0_mps^2/g_mps2)*sind(2*theta_deg) - target_m;
    slope_m_per_deg = (v0_mps^2/g_mps2)*cosd(2*theta_deg)*(2*pi/180);
    if abs(f_current_m) < tolerance_m
        break
    end
    theta_deg = theta_deg - f_current_m/slope_m_per_deg;
end
newton_root_deg = theta_deg;
```

Responsible-AI caution. A generated Newton script can be runnable but still use a radians derivative with a degrees residual, or it can reverse the residual sign without explaining it. Check the physical residual, the derivative unit, the update trace, and the final substitution.

7. Validate by substitution

A root is not accepted because a loop stopped. Substitute the computed angle back into the residual:

$$R(\theta_*) = \frac{20^2}{9.81}\sin(2\theta_*), \qquad f(\theta_*) = R(\theta_*) - 35.$$

For the low-angle root,

$$\theta_* \approx 29.6^\circ, \qquad |f(\theta_*)| < 10^{-3}\,\text{m}.$$

The root is an angle. The residual is a distance. Keeping these units separate prevents a common interpretation error.

8. Diagnose a bad bracket

A midpoint can always be calculated, but bisection should not start unless the two endpoint residuals have opposite signs. If both endpoints undershoot or both overshoot, the interval does not provide the required bracket evidence for a root.

Check	Why it matters
Graph crosses zero	gives a visual root location
Endpoint signs differ	gives a numerical bracket for bisection
Residual after solving is small	validates that the computed value satisfies the target condition
Root unit is stated	preserves the physical meaning of the answer

Guided extension: comparing iteration effort

Use the supplied bisection and Newton results to read how the two methods behave. Bisection keeps a bracket and is easy to defend. Newton often needs fewer iterations, but it depends on a useful local slope and a reasonable starting value. Treat iteration count only as a limited measure of computational effort; it never replaces residual validation.

Optional extension

Uniqueness proof, multiple stopping criteria, MATLAB `fzero`, safeguarded Newton, and formal method selection are optional and are not required. The Week 5 required pathway only needs one meaningful residual, one valid bracket, starter bisection and Newton traces, one stopping rule, and substitution validation.

Three takeaways

1. Convert the physical target condition into a residual equation $f(x) = 0$ before solving.
2. Bisection uses sign-changing brackets; Newton uses a local tangent update.
3. A computed root must be validated by substituting it back into the residual and interpreting its physical unit.

Preparation for the practical. Be ready to transfer the same root-finding logic to a projectile target, a thin-lens image distance, and an RC threshold time. For every context, state the residual, unknown unit, bracket or starting value, stopping rule, validation check, and physical meaning of the root.